



Ultra-Low Latency AI for Real-Time Decision Making

Built for the Cerebras Ultra-Low Latency AI Track

The Problem

The Latency Problem

- Traditional AI systems rely on batch inference
- Even small delays break real-time decision loops
- Interactive AI feels slow when latency exceeds milliseconds

Latency → Loss of Responsiveness

The Solution

Project Surge

- Event-driven AI inference engine
- Designed for sub-millisecond response times
- Powered by Cerebras ultra-low-latency inference

Latency is treated as the primary design constraint.

How It Works

Real-Time Inference Pipeline

Event Stream → Cerebras Inference → Immediate Action

$$y_t = f_{\theta}(x_t), t < 1 \text{ ms}$$

Raindrop Learning Layer

Continuous Learning with Raindrop

- Learns from incoming events in real time
- Adapts responses based on outcomes
- Improves accuracy without slowing inference

$$\theta_{t+1} = \theta_t + \alpha \nabla L$$

Why Cerebras

Why Cerebras?

- Deterministic ultra-low latency
- Massive parallelism without batching
- Predictable performance under load

Project Surge shows what becomes possible when AI latency approaches zero.